



The impact of uncertainty shocks in emerging economies



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ABSTRACT

A recent strand of research proposes that sudden jumps in uncertainty generate rapid drops and recoveries in real macroeconomic variables that drive the business cycle. Using an empirical model, we find substantial heterogeneity in reactions to these shocks across countries. In comparison to the U.S. and other developed countries, emerging economies suffer much more severe falls in investment and private consumption following an exogenous uncertainty shock, take significantly longer to recover, and do not experience a subsequent overshoot in activity. We provide evidence that the dynamics of investment and consumption are correlated with the depth of financial markets. We control for the potential role of a credit channel, and estimate that it can account for up to one-half of the increased fall in investment generated by uncertainty shocks among emerging economies with less-developed financial markets. In this context, monetary and fiscal policy actions that alleviate the impact of credit constraints facing firms may reduce the impact of uncertainty shocks in these economies.

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1. Introduction

Sudden changes to the level of aggregate uncertainty facing economic agents have been shown to be an important shock driving the U.S. business cycle. Using a simple reduced-form VAR, [Bloom \(2009\)](#) estimates that U.S. industrial production is reduced by approximately 1% in response to an uncertainty shock. The initial drop is followed by a swift recovery and subsequent overshoot in production that surpasses its trend by approximately 1%. [Gourio et al. \(2013\)](#) find that similar dynamics are observed in a group of G7 countries following spikes in global uncertainty. The main purpose of this paper is to examine to what extent these findings generalize to a broader group of countries. We estimate the response of investment and private consumption to global uncertainty shocks in 40 countries using an open-economy VAR approach, where our shock corresponds to strong increases in the option-implied uncertainty of the U.S. stock market.

We find that the impact of uncertainty shocks varies across countries. The evidence we present for a large group of developed economies is consistent with the literature, generating a rapid drop and rebound in investment with private consumption nearly unaffected. These dynamics are consistent with the predictions of a model with fixed investment costs that generate a real-option value of waiting under uncertainty, as stressed by [Bloom \(2009\)](#). In contrast, the evidence we present for emerging markets indicates that these economies suffer a much more

severe fall in investment and that this fall is considerably more persistent. We also observe a significant fall in private consumption following global uncertainty shocks, in contrast to developed economies where consumption deviates very little from its trend.

It has been proposed that uncertainty generates reductions in real activity since at least [Keynes \(1937\)](#), who suggested that investment is the most volatile component of aggregate demand precisely because it relies most heavily on opinions about future events, which are necessarily ill-informed. Building on work by [Weisbrod \(1964\)](#) and [Arrow and Fisher \(1974\)](#), [Bernanke \(1983\)](#) formalized the idea that, when projects are irreversible and information is made available over time, the presence of uncertainty about future returns generates an option value of waiting that lowers the rate of current investment, even when agents are risk-neutral.¹ Recent work on the topic has made progress on modeling the effects of uncertainty, on verifying the predictions of the theoretical models with their empirical counterparts, and on estimating the economic significance of the mechanism.

In models of investment with fixed adjustment costs, such as [Bloom \(2009\)](#), the region of inaction – in which firms find it optimal not to adjust their input levels – varies according to the level of time-varying uncertainty. In periods of high uncertainty, more firms choose to “wait-and-see”, putting their investments on hold voluntarily. When uncertainty dissipates and business conditions can be better ascertained, many firms who postponed their factor adjustments find

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¹ See [Pindyck \(1991\)](#) and [Dixit \(1992\)](#) for an overview of the implications of irreversibility on investment under uncertainty.

themselves far from their optimal levels of capital and labor, and carry out the adjustment required to relieve their pent-up factor demand. This generates a rapid recovery and overshoot from the original trend levels of macro variables. One prediction of this modeling approach is that a given fiscal or monetary policy intervention will have a smaller impact on real activity during periods of high uncertainty.

A number of candidate explanations are available to account for the heterogeneity in responses we observe across countries, which lead to very different policy prescriptions in the face of uncertainty shocks. While the model described above could be calibrated according to local depreciation and discount rates, the evidence from emerging market economies suggests that other factors need to be taken into account to fully characterize the response of real activity to uncertainty shocks. We explore several alternative mechanisms using a cross-sectional approach in order to shed light on the heterogeneity in response dynamics, but do not offer conclusive evidence in this paper. We document that the amplitude of the fall in investment and private consumption for the group of countries in our sample is correlated with GDP-per-capita, the depth of local financial markets, an index of business-related institutional quality, and the degree of financial dollarization.

Following the literature, we note that credit constraints affecting firms and households due to characteristics of local financial markets are relevant features of emerging economies. We conduct a counterfactual exercise to gauge the importance of the credit channel in amplifying the real impact of uncertainty shocks. Among emerging economies with less-developed financial markets, we find that half of the increased fall in investment can be accounted for by the drop in credit. In other words, when we shut off the endogenous response of credit to the uncertainty shock, we observe that investment dynamics in emerging economies more closely resemble those in developed economies.

Our study is related to the literature that has documented the relationship between volatility and the business cycle. Fernández-Villaverde et al. (2011) study how changes in the volatility of the real interest rate at which emerging economies borrow impact real variables in the economy. Using an estimated stochastic volatility process for real interest rates in a standard small open economy business cycle model calibrated to match data for Argentina, Brazil, Ecuador and Venezuela, they show that an increase in real interest rate volatility triggers a fall in output, consumption, investment, and hours worked, and a notable change in the current account.

Consistent with our interpretation of uncertainty shocks interacting with credit constraints in emerging markets, Matsumoto (2011) studies the impact of shocks to global liquidity conditions on a group of emerging economies. The availability of global liquidity for Argentina, Brazil, Chile and Mexico is measured using the VIX index. He shows that positive global liquidity shocks in emerging market economies tend to generate lower interest rates, more appreciated exchange rates, higher output and higher equity returns.

Beyond its effect on the business cycle, volatility has also been documented to have an impact on long-run growth rates. Using a panel of 92 countries for the period of 1960 to 1985, Ramey and Ramey (1995) document a strong negative relationship between output growth and the standard deviation of per capita annual growth rates over time. They explain that “...the negative effect of volatility stems mainly from volatility of innovations to GDP growth, which reflects uncertainty.”

The paper is organized as follows. In Section 2 we present the data used in the paper and discuss our definition of a global uncertainty shock. In Section 3 we describe our methodological approach, and state the econometric model we will estimate. In Section 4 we present the results of our estimations, examine the heterogeneity in the response functions across countries using plausibly relevant variables, and discuss the policy implications of each interpretation. We then conduct robustness checks using alternative definitions of the uncertainty shock and specifications of the model. Section 5 concludes.

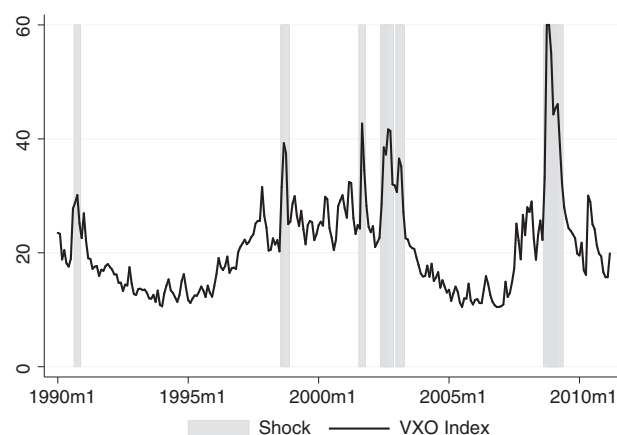


Fig. 1. U.S. volatility index and global uncertainty shocks.

2. Data

While the idea that uncertainty is an important determinant of the business cycle has been acknowledged since Keynes (1937), its incorporation into modern models of the business cycle was delayed due to its intangible nature. There has been disagreement as to what, exactly, “uncertainty” corresponds to, and what its empirical counterpart should be. In an important contribution, Bloom (2009) showed that measures of implied and observed stock-market volatility are strongly correlated with other measures of both micro- and macro-level uncertainty, including bond spreads, disagreement among professional forecasters, and the distributions of firm profits and industry productivity growth.² This has motivated the use of implied volatility derived from stock market options as a measure of forward-looking aggregate uncertainty.

In measuring uncertainty for a panel of countries, it is unclear whether the relevant measure is a local indicator of volatility, or a common global shock that would affect all economies exogenously. Gourio et al. (2013) find that country-level risk indices constructed using local financial indicators are highly correlated across countries. Given this extensive co-movement, they use the mean of their country-level indices as a common measure of global uncertainty in their analysis.

Since reliable high-frequency domestic series are not always available for emerging market countries, a measure built using averages of these domestic series would be susceptible to changes in the sample of countries included. It is also debatable to what extent stock market indices are an adequate measure of local business conditions in an emerging economy, especially during the 1990s when market capitalization was small relative to output and very few firms sought financing in local stock markets. For this reason, we use a single series to identify global shocks in our core analysis, and conduct a sensitivity exercise in Section 2 to test whether our results are robust to using uncertainty series built with domestic financial variables for emerging markets.

Our measure of uncertainty is the VXO index, which corresponds to the volatility implied by a synthetic 30-day option on the S&P 100 stock index. We label the variable vxo_t and plot the series over our sample period in Fig. 1. We then implement the methodology presented in Bloom (2009) to identify global uncertainty shocks as periods of spikes in vxo_t . Uncertainty shocks are identified as periods in which the Hodrick–Prescott de-trended $\hat{vxo}_t$ exceeds its mean by more than 1.65 standard deviations. In Fig. 1, identified shocks correspond to the shaded regions. Six events that fit this criteria are

² Bloom et al. (2007) had previously shown that share return volatility is correlated with firm-level uncertainty in the United Kingdom.

Table 1
Sample of countries.

Developed	Emerging
Core sample (1990:1 to 2011:1)	
Australia	Japan
Austria	Netherlands
Canada	New Zealand
Denmark	Norway
Finland	Switzerland
France	United Kingdom
Italy	United States
Extended sample (199* to 2011:1)	
Belgium	Argentina
Israel	Brazil
Germany	Colombia
Russia	Croatia
Spain	Czech Republic
Sweden	Estonia
	Hungary
Total developed: 20	Total emerging: 20

identified in our sample period, and each can be clearly associated with an important geopolitical or financial event that can be considered exogenous to local macroeconomic fundamentals.³ The persistently high volatility following the 2008 financial crisis has increased the standard deviation of the series as a whole, such that not all events reported by Bloom (2009) are identified as shock episodes even though we employ the same identification criteria. In particular, the 1997 Asian financial crisis is no longer identified as a global uncertainty event.⁴

The events are weighted according to the magnitude of the volatility shock, making the results comparable across countries even though the events under consideration may differ according to the length of each sample. When volatility remains high for more than one quarter, as is the case in the Worldcom/Enron scandals of 2002 and the global financial crisis of 2008–09, the vol_t^* variable maintains a positive value until volatility drops below the threshold. Explicitly, we define the global uncertainty shock series as the following split function:

$$vol_t^* = \begin{cases} 0 & \text{if } v\tilde{x}_t \leq \text{threshold,} \\ v\tilde{x}_t & \text{if } v\tilde{x}_t > \text{threshold.} \end{cases}$$

The quarterly series is then constructed such that a quarter containing a shock is assigned the arithmetic mean of $v\tilde{x}_t$ over the underlying months, with others receiving a zero value. There are many alternative ways in which the shock series could be constructed, and we carry out robustness checks using alternative definitions of the volatility variable in Section 4.

Local macroeconomic variables are from the quarterly national accounts compiled by the Organization for Economic Cooperation and Development, the International Monetary Fund's *International Financial Statistics*, and in some cases are supplemented by publicly-available data from national central banks and statistics agencies. Series reported in volumes are used when available, and we deflate nominal series using the national consumer price index to obtain constant-price series in the remaining cases. All series are seasonally adjusted using the X-12-ARIMA routine provided by the U.S. Census Bureau, and detrended using the Hodrick–Prescott filter.⁵

³ Interestingly, many of the events identified as uncertainty shocks using this criteria coincide with those listed in Caballero and Krishnamurthy (2008) as episodes of increased Knightian uncertainty that generate a flight to quality event.

⁴ The Asian crisis is included as a robustness check in Section 4.

⁵ While most series appear stationary in first-differences, this transformation is particularly important in the estimation of the model for countries that have undergone periods of very high inflation, such as Argentina, Mexico, Peru and Turkey.

The full list of countries included in our analysis is displayed in Table 1. All the countries in our core sample have complete data for the period of 1990:1 to 2011:1. We use a common sample period to ensure that results are comparable across countries, since the shock we are considering is common to all countries and the structure of the international financial system has changed substantially over time. Countries in the extended sample, for which we have data starting between 1991:1 and 1997:1, are included later as robustness checks and in the cross-section correlation exercises.

3. Empirical specification

We use a standard specification for the reduced-form vector autoregression that incorporates stock market levels, the volatility shock, prices and real variables. The full set of variables included in the VAR for each country are: the cyclical component of the S&P-500 index, $\tilde{s}p_t^*$; the weighted uncertainty shock indicator, vol_t^* ; the cyclical deviation of consumer price inflation, $\tilde{\pi}_t$; and the cyclical deviations of real investment and private consumption, $\tilde{I}_t$ and $\tilde{C}_t$. Two lags of each variable are included, and all data are at a quarterly frequency.

For our estimation of the United States, we allow the stock returns and volatility variables described in Eqs. (1) and (2) to be affected by lags of the endogenous vector $\mathbf{X}_t$. For the other countries in our sample, we impose a small open economy assumption such that S&P stock returns and the uncertainty variable are exogenous to the local economy. The reduced form specification for those countries can thus be written as:

$$\tilde{s}p_t^* = \rho_1 + \beta_1(L + L^2)\mathbf{Z}_t + \varepsilon_t^{sp*} \quad (1)$$

$$vol_t^* = \rho_2 + \beta_2(L + L^2)\mathbf{Z}_t + \varepsilon_t^{vol*} \quad (2)$$

$$\tilde{\pi}_t = \rho_3 + \beta_3(L + L^2)\mathbf{Z}_t + \gamma_3(L + L^2)\mathbf{X}_t + \varepsilon_t^\pi \quad (3)$$

$$\tilde{I}_t = \rho_4 + \beta_4(L + L^2)\mathbf{Z}_t + \gamma_4(L + L^2)\mathbf{X}_t + \varepsilon_t^I \quad (4)$$

$$\tilde{C}_t = \rho_5 + \beta_5(L + L^2)\mathbf{Z}_t + \gamma_5(L + L^2)\mathbf{X}_t + \varepsilon_t^C \quad (5)$$

where $\beta_1, \dots, \beta_5, \gamma_3, \dots, \gamma_5$ are vectors of parameters, $\mathbf{Z}_t = \{\tilde{s}p_t^*, vol_t^*\}$ is a vector of exogenous shocks, and $\mathbf{X}_t = \{\tilde{\pi}_t, \tilde{I}_t, \tilde{C}_t\}$ is a vector of endogenous variables. The inclusion of the stock market returns in Eq. (1) is meant to control for first-moment shocks to returns, such that our analysis of shocks to uncertainty can be interpreted as the impact of the uncertainty shocks alone.

The model is estimated by maximum likelihood for each of the countries in our sample. We then compute the impulse-response matrix Φ_i and its orthogonalized counterpart Θ_i using a Cholesky decomposition of the matrix $\hat{\Sigma}$, which contains estimates of the contemporaneous covariance between the reduced-form error terms. In the familiar notation employed by Lütkepohl (2006), the orthogonalized impulse-response functions are given by

$$\Theta_i = \Phi_i \cdot \mathbf{P}, \quad \text{where } \hat{\Sigma} = \mathbf{P}\mathbf{P}'.$$

As in Bloom (2009), our main identifying assumption is that the uncertainty shock is predetermined with respect to variables from the local economy. As discussed in Section 2, the variable corresponds to spikes in the VXO index, which captures the option-implied volatility of the S&P 100 index. While it may be argued that the continuous volatility series depends on stock market levels, and thus on other measures of economic activity, the shocks to this series shown in Fig. 1 can all be associated with important geopolitical events, and are thus quite exogenous even to U.S. macroeconomic variables.

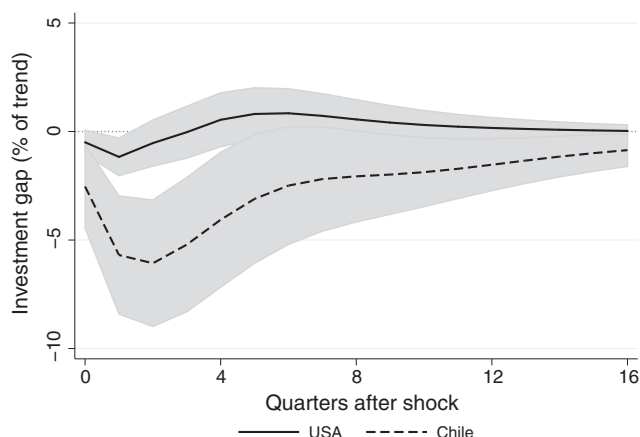


Fig. 2. Response of investment to an orthogonalized global uncertainty shock.

The magnitude of the orthogonalized shock to uncertainty is equal to the mean value of v_{x0t} over all episodes in our sample. The response functions have been normalized to percentage units such that the plots can be interpreted as the response variable's deviation from trend in percentage points following a volatility shock of average magnitude.

4. Results

4.1. Main findings

We estimate the model separately for each of the countries listed in Table 1. Fig. 2 displays the orthogonalized impulse response functions for investment in Chile and the United States following a shock to global uncertainty, with one-standard-deviation confidence intervals. Standard deviations have been estimated using a parametric bootstrapping procedure with 200 repetitions. While the U.S. reaction is consistent with the results in Bloom (2009) and Gourio et al. (2013) – displaying a rapid drop, rebound, and overshoot – the Chilean case displays strikingly different characteristics. First, the amplitude of the drop in investment is many times larger. Second, investment takes considerably longer to return to its trend level, indicating that the effect of the uncertainty shock is more persistent. Third, the response of investment in Chile displays no subsequent overshoot.

To generalize our findings, we group the countries into two categories: developed and emerging. We do so using each country's World Bank income classification at the beginning of the sample period (1990), where the “developed” group corresponds to high-income countries and the “emerging” group to all other countries (World Bank, 1990).⁶ Fig. 3 plots the median impulse response functions for the group of emerging market economies and for the group of developed countries in the core sample.⁷ The first panel reports the response of investment. The response for developed countries is consistent with the response estimated for the United States, displaying a rapid drop and rebound in investment. The group of emerging economies suffers a much more severe fall in investment following the global uncertainty shock, the median gap falling by four times the fall experienced in developed economies. This result is consistent with Lane's (2003) finding that the business cycle in emerging markets is considerably more volatile than in wealthier countries. We also observe considerable heterogeneity

in the persistence of the shock across countries. In general, emerging markets take substantially longer to recover to their pre-shock trend, and display no subsequent volatility overshoot.

The second panel shows the response functions for private consumption. While consumers in developed countries are able to smooth their consumption and thus avoid a drop in utility due to the uncertainty shock, private consumption in emerging markets falls substantially and persistently, returning to trend after five quarters. Possible explanations for this difference are a constraint on consumer access to financial markets, or to differences in the social safety nets in place in each country. The drop in private consumption is consistent with Bernanke (1983), who argues that consumption in durable goods should fall along with business fixed investment during periods of high uncertainty, since it is subject to a similar degree of irreversibility that would lead agents to postpone their purchasing decisions until uncertainty had subsided.⁸ The fall in private consumption observed in emerging economies suggests that consumers are likely suffering a transitory loss of welfare as a result of the uncertainty shock.

4.2. Potential explanatory channels and their policy implications

In models that feature fixed adjustment costs and an option-value of waiting mechanism, firms reduce their investment voluntarily during a period of increased uncertainty as their region of optimal inaction widens. The contraction is independent of changes in the expected value of future demand, and takes place even when firms are risk neutral. An implication, which has been explored since Dixit (1992), is that firms are less responsive to a given change in demand or cost conditions during periods of high uncertainty. Bloom et al. (2007) employ a panel of British firm-level data to estimate the degree to which aggregate and idiosyncratic uncertainty reduce the responsiveness of investment to demand shocks. They find that firms in the fourth quartile of uncertainty are half as responsive as those in the first quartile, suggesting that the effect of uncertainty is of first-order importance to the firm's investment decision. This leads to the conclusion that both monetary and fiscal policy interventions are ineffective at maintaining rates of investment during episodes of high uncertainty.

But can this class of models be calibrated to generate the observed heterogeneity across countries that we have reported above? The value of waiting is influenced by the size of the fixed cost associated with investing – or, equivalently, the degree of irreversibility of capital – faced by firms, which is likely to vary across countries. Pindyck (1991) discusses how the degree of irreversibility in an economy is determined in part by the market environment, such as regulations on capital mobility and the risk of expropriation. The sunk costs associated with filing lengthy paperwork and overcoming bureaucracy also increase the irreversibility of projects. Countries with regulatory frameworks that make investments less reversible should thus generate larger real-option values to waiting during periods of uncertainty, and thus suffer deeper recessions.

Another parameter that determines the value of waiting in the model is the stochastic discount factor that firms use to discount future profits. When the future is heavily discounted, the value of putting off current projects in order to wait for stability to return goes down, and firms will be more likely to go ahead with projects despite high levels of uncertainty. Cross-country differences in discount rates could thus account for some of the variability in amplitude.⁹

⁶ Although it is (just barely) classified as a high-income country, we include Hong Kong in the group of emerging market economies.

⁷ The value of the median IRF in step i is defined as the median across all IRFs in step i , and thus does not correspond to the complete IRF of a single representative country. The confidence interval for emerging markets corresponds to the standard deviation of the median country used in step i . Medians are used as our measure of central tendency to ensure that results are not influenced by the presence of outliers.

⁸ Carrière-Swallow and Medel (2011) carry out a sectorial analysis for Chile and confirm that the fall in private consumption following an uncertainty shock is due to a very large fall in consumption of durable goods.

⁹ A real depreciation may increase the amplitude of the fall in investment due to the fact that the import content of investment is high in some emerging economies, as documented by Burstein et al. (2004).

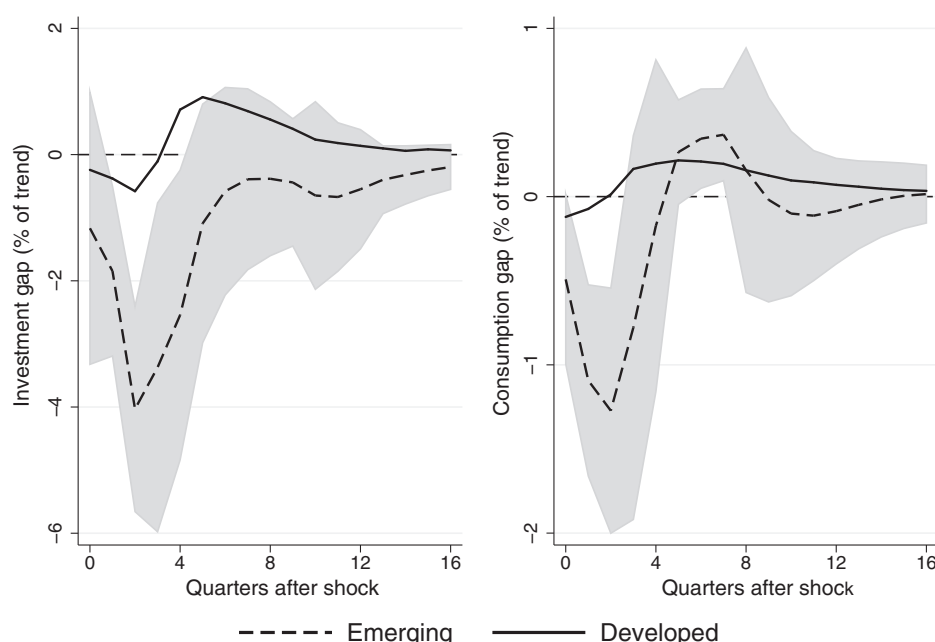


Fig. 3. Response of investment and private consumption to an orthogonalized global uncertainty shock.

While these parameters can help describe the heterogeneity in amplitude, they cannot directly account for the slow rate of recovery or for the absence of an overshoot in the medium-run. The real-options mechanism generates a fall in investment in response to increased uncertainty by widening the region of inaction, and subsequent recovery to trend once the uncertainty returns to normal levels. In Bloom (2009), the overshoot estimated for the U.S. economy is modeled using a separate mechanism. The increased volatility following the uncertainty shock, when combined with a positive rate of depreciation, skews the distribution of firms in capital-to-productivity space, creating a glut of firms on the edge of their inaction regions that initiate investment projects once the level of uncertainty has fallen back to normal levels. This “volatility overshoot” generates a positive response in the medium-term, and once uncertainty has fallen sufficiently, is enough to generate the overshoot in the reaction function. The shape of the impulse response function for emerging markets is similar to that reported by Bloom in his decomposition exercise as the result of the uncertainty channel alone, suggesting that volatility effects might not have the same effect in emerging markets.

In order to examine the differences in the investment reaction across countries, we compute for each country the amplitude of each impulse-response function for the investment gap Eq. (4) following an orthogonalized shock to uncertainty. Then, we compute the correlations between the amplitudes and potentially relevant country

characteristics. Amplitude is defined as the smallest value of the investment or consumption reaction function following the shock (in other words, Chile is assigned a value of -6%). For this exploratory exercise, we estimate the model for the extended sample of countries – including those with data start dates up to 1997:1 – in order to increase the number of data points. The country-specific variables are evaluated at their 2004 levels (Table 2).

There is a positive correlation between the amplitudes of the fall in investment and consumption and a country's GDP-per-capita in 1990, generalizing the finding that wealthier countries suffer smaller falls in these variables following uncertainty shocks. As a proxy for the fixed costs or investment irreversibility in each country, we use the World Bank's *Doing Business* index, which measures the degree to which a country's regulatory framework is friendly to business.¹⁰ The negative correlation suggests that countries with lower-quality institutions experience deeper falls in investment following global uncertainty shocks.

In this analysis we have estimated the responses of investment and private consumption across countries to a common uncertainty shock. While difficult to measure, it is possible that the same episode of high global uncertainty may be amplified by local policy, such that the level of uncertainty actually perceived by economic agents differs between countries. Pindyck (1991) warns that “if a goal of macro-economic policy is to stimulate investment, stability and credibility may be more important than the particular levels of tax rates or interest rates.” Baker et al. (2011) construct an index of U.S. policy uncertainty using Google News data on media coverage of related keywords, and estimate that shocks to this index have first-order effects on investment behavior. The sharp falls we observe in EMEs may be caused by the magnification of global uncertainty shocks due to a local environment of elevated policy uncertainty. The sensitivity exercise performed by Bloom (2009) suggests that the magnitude of the uncertainty shock could account for the amplitude of the fall in investment, but would also be accompanied by an equally magnified overshoot, which is not observed in our estimations.¹¹

Unfortunately, the parameters required to improve country-level fit in the adjustment costs model are difficult to measure, making it

Table 2
Summary statistics across countries: Response function amplitudes.

	Investment	Private consumption
Developed economies		
Median	−0.82	−0.21
IQR	[−1.30, −0.08]	[−0.51, −0.09]
Emerging economies		
Median	−4.79	−1.29
IQR	[−6.50, −1.62]	[−2.09, −0.60]
Correlations with:		
GDP per capita	0.61	0.57
Doing Business	−0.36	−0.31
Financial depth	0.27	0.31
Dollarization	0.04	−0.41

Notes: (i) Amplitude is defined as the smallest value in the country's IRF following an uncertainty shock; (ii) variables correspond to 2004 values; (iii) correlations computed across the 40 countries listed in Table 1.

¹⁰ Note that a lower score indicates better institutional quality.

¹¹ See Bloom (2009), p. 672 for a discussion of this sensitivity exercise.

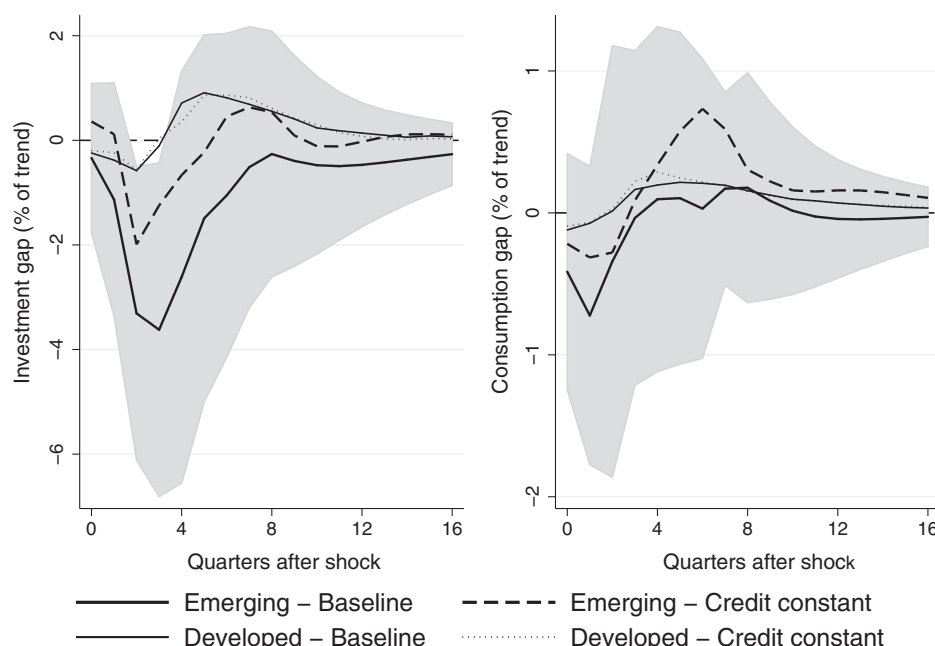


Fig. 4. Response of investment and private consumption to an orthogonalized global uncertainty shock.

difficult to establish to what extent the impact of the uncertainty shocks are being generated by voluntary reductions in firm investment, or by some other mechanism.

A plausible explanation that could account for the differences in the evolution of investment after an uncertainty shock is the presence of financial frictions. As the economy enters a period of high uncertainty, firms may find it more difficult to obtain financing for their projects if (i) banks and other financial intermediaries find it more difficult to gauge the degree of risk involved in the project, (ii) banks are unable to obtain external financing themselves due to a shortage of liquidity or flight-to-quality episode as described by Calvo and Mendoza (2000), or (iii) firms suffer a deterioration of their balance sheets – perhaps due to a currency mismatch and sudden depreciation – which reduces the collateral available to post against new loans. The fall in the collateral value and/or the deterioration in the firms' balance sheets increase the negative effect of the uncertainty shock in the economy both in terms of the initial fall but also in terms of the persistence of the drop in investment.

An extensive literature has explored the link between emerging market recessions and the functioning of financial markets. Claessens et al. (2011) carry out a comprehensive business-cycle analysis of recessions and recoveries for a sample of 45 countries. One of their findings is that recessions in emerging market countries are more often accompanied by financial market disruptions than is the case in developed economies. Aizenman and Powell (2003) show that a weak legal system combined with high information verification costs generates large, first-order effects of volatility on production, employment and welfare.

It is important to indicate that the relationship between the depth of financial markets and real volatility in emerging markets may not be linear. Aghion, Bacchetta and Banerjee (2004) use a dynamic open economy model with financial frictions to study the role of financial factors as a source of instability in small open economies. Their assumption is that the amount firms can borrow is limited to μ times the amount of their current investible funds. Their analytical result is that economies at an intermediate level of financial development – rather than the very developed or the severely underdeveloped – are the most unstable. Thus, in our analysis emerging market economies should exhibit more volatility than developed economies and less developed economies.

In another study, Neumeier and Perri (2005) document that in emerging economies real interest rates are more volatile, strongly

countercyclical and lead the cycle. They construct a simple dynamic general equilibrium model of a small open economy where shocks to the real interest rate are amplified through the presence of working capital constraints. They show that these considerations are useful to explain the business cycle in emerging market economies. Uribe and Yue (2006) also highlight the role played by working capital constraints in the amplification of shocks hitting an emerging market economy.

Another factor that may generate amplification in the presence of financial frictions is the accumulation of liabilities in foreign currency. Céspedes et al. (2004) point out that financial frictions in emerging markets, and their interaction with incurring liabilities in foreign currency, present an important shock-amplification mechanism that magnifies the business cycle in these countries. As a measure of the importance of foreign-currency-denominated debt in the local economy, we use the degree of dollarization reported by Levy Yeyati (2006). While there is no correlation between amplitude of investment and dollarization, there is a strongly negative correlation between the reaction of consumption and dollarization. This suggests that countries indebted in a foreign currency suffer deeper falls in consumption following uncertainty shocks, possibly because balance sheet effects of currency devaluations cause an external consumer credit crunch.

A firm credit constraint could affect (i) the extensive margin, by reducing firms' ability to maintain operations while postponing profitable projects during the period of uncertainty, and (ii) the speed at which firms can satiate pent-up demand once the shock has dissipated, and thus the duration of the episode. We find that there is some correlation between IRF amplitudes and certain variables used in the literature to account for credit constraints. As a proxy of a country's financial depth, we use the mean ratio of private credit to GDP over the full length of the period under analysis. There is a positive – though weak – correlation between the reaction of investment and this measure of financial depth, suggesting that countries with shallower financial markets experience somewhat deeper falls in investment following global uncertainty shocks.

If financial frictions are in fact responsible for a substantial portion of the impact of uncertainty on investment, the policy implications are quite different to viewing the downturn as a voluntary and optimal decision to delay investment. If firms in fact desire to invest, but are unable to obtain financing due to imperfections in the financial market, then

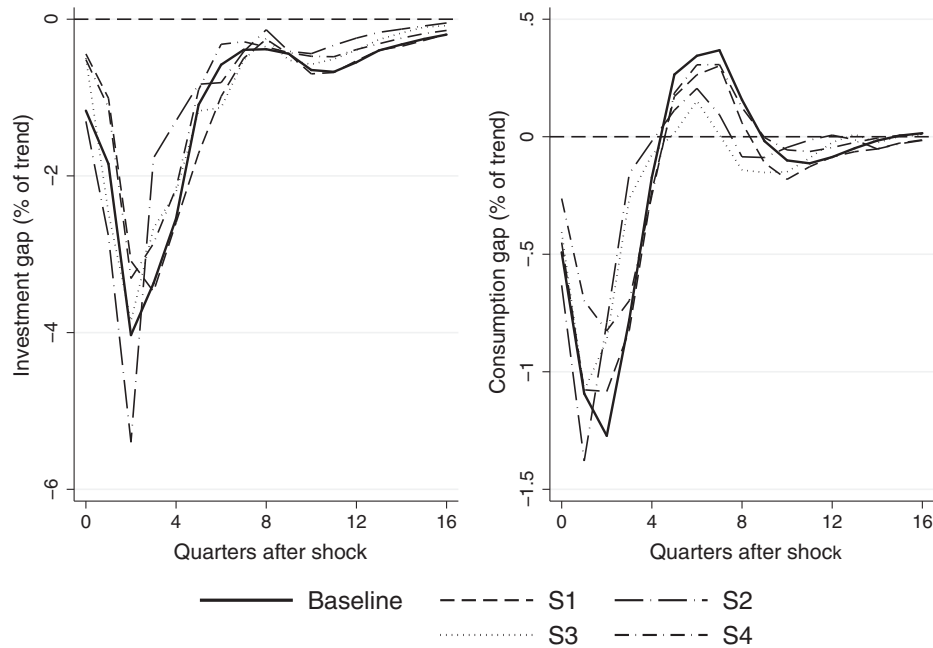


Fig. 5. Impulse-response functions following an uncertainty shock using alternative shock variables.

liquidity operations by the monetary authority could potentially reduce the amplitude of the fall and increase the speed of recovery once uncertainty has dissipated.

Interestingly, the duration of the drop in investment is not correlated with any of our four explanatory candidates. This may not be surprising as we are considering a limited set of variables to explain the heterogeneity in our simple exercise, and a combination of these and other structural characteristics are likely responsible for the duration of recoveries.

4.3. The credit channel

A possible amplification mechanism and source of persistence for the uncertainty shock is the credit channel. As [Matsumoto \(2011\)](#) proposes, a rise in the VIX index – a close cousin of the VXO that forms the basis of our measure of uncertainty – is associated with reduced global liquidity and availability of credit in emerging economies. In a first attempt at measuring the importance of this channel, we carry out a counterfactual exercise to answer the question “What would have been the impact of the uncertainty shock in emerging markets, if there had been no change in the flow of credit?” To do so, we employ a methodology proposed in a preliminary version of [Sims and Zao \(2006\)](#) and later used by [Bernanke et al. \(1997\)](#) to identify the direct impact of exogenous shocks to the price of oil in the presence of an endogenous monetary policy response.

We augment the reduced form specification with an additional equation to model the dynamic of the country's loans by banks, which is assumed to be subject to the external volatility shock and endogenously determined along with other local macro variables. The augmented reduced form is thus:

$$\begin{aligned} \tilde{s}p_t^* &= \rho_1 + \beta_1(L + L^2)Z_t + \varepsilon_t^{sp} \\ vol_t^* &= \rho_2 + \beta_2(L + L^2)Z_t + \varepsilon_t^{vol} \\ \tilde{\pi}_t &= \rho_3 + \beta_3(L + L^2)Z_t + \gamma_3(L + L^2)X_t + \varepsilon_t^{\pi} \\ \tilde{\Delta}B_t &= \rho_4 + \beta_4(L + L^2)Z_t + \gamma_4(L + L^2)X_t + \varepsilon_t^B \\ \tilde{l}_t &= \rho_5 + \beta_5(L + L^2)Z_t + \gamma_5(L + L^2)X_t + \varepsilon_t^l \\ \tilde{c}_t &= \rho_6 + \beta_6(L + L^2)Z_t + \gamma_6(L + L^2)X_t + \varepsilon_t^c \end{aligned}$$

where $\tilde{\Delta}B_t$ is the cyclical deviation of the growth rate of domestic bank loans to the private sector, and $X_t = \{\tilde{\Delta}B_t, \tilde{\pi}_t, \tilde{l}_t, \tilde{c}_t\}$ is the augmented vector of endogenous variables.

A reduction in loans after an uncertainty shock could simply reflect a fall in loan demand, and is therefore indistinguishable from voluntary wait-and-see behavior. Nevertheless, shutting off the dynamics associated with the loan variable gives us an idea of how much quantitative relevance the credit channel could have on investment and consumption dynamics in the face of high uncertainty. Our counterfactual places restrictions on the matrix of estimated contemporaneous correlations, $\hat{\Sigma}^R$, and on the estimated parameters. By setting all elements in the row and column corresponding to the credit channel to zero, we simulate an environment where credit does not respond contemporaneously to shocks from the other variables in the system. When constructing the restricted impulse-response matrix, Φ_t^R , we set all components of the vectors $\hat{\beta}_4$ and $\hat{\gamma}_4$ to zero, such that credit does not respond to lagged values of other variables either. The orthogonalized impulse-response functions are then constructed by multiplying Φ_t^R by the Cholesky decomposition of $\hat{\Sigma}^R$. By imposing the restrictions after estimating the reduced form, we ensure that the estimations do not suffer from omitted variable bias.

We estimate this model for the core sample of developed economies, and a sample of countries with relatively less-developed financial markets.¹² The results are shown in [Fig. 4](#). The baseline results are shown by the solid lines, while the credit counterfactuals are shown by dashed lines. Controlling for the credit channel changes the dynamics of the recovery in these emerging economies. The increased amplitude of the fall in investment is reduced by almost half, from -3.6% to -2.0% , compared to a fall of -0.6% in developed countries. The recovery period for these emerging markets is also shortened from 9 quarters to 6. Strikingly, controlling for credit seems to make almost no difference among developed countries, which is consistent with the hypothesis that financial frictions play a limited role at the aggregate level for those countries during uncertainty shocks. However, the confidence interval associated with this more heterogeneous group of countries is

¹² The emerging economies included in the exercise are Argentina, Chile, Colombia, the Czech Republic, Malaysia, Mexico, Turkey and South Africa. The estimation period is 1990:1 to 2011:1, which is the same as has been used in the benchmark results presented so far.

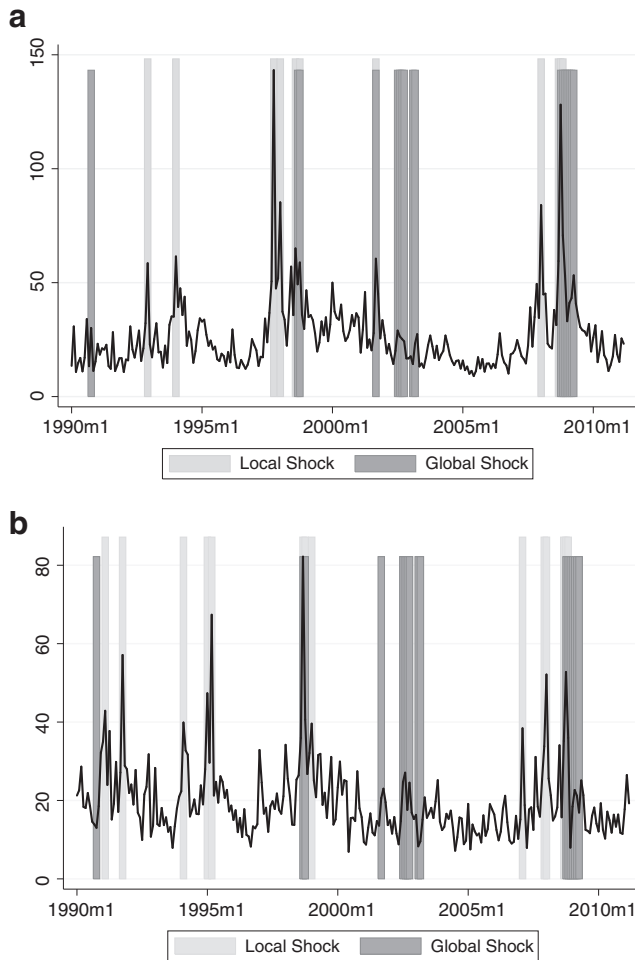


Fig. 6. Realized return volatility in Hong Kong and Chile.

wider than in our benchmark sample, and the results of the counterfactual exercise do not appear statistically significant.

We should point out that the distinction between developed and emerging economies is somewhat arbitrary, and the importance of the credit channel varies within the group of emerging economies. In Hong Kong, South Korea and the Philippines, for instance, we estimate that the credit channel plays a negligible role in amplifying uncertainty shocks. In the first two, this is likely due to the presence of highly developed financial markets. In the latter case, very poor institutions may make firms less reliant on formal lenders across the cycle.¹³

These results suggest that the large impact of uncertainty shocks on investment and consumption in many emerging markets is due in part to tightening credit constraints. Once we hold the flow of credit constant, these emerging economies display dynamics that more closely resemble those of developed countries, and which are easier to reconcile with Bloom's (2009) modeling approach. We acknowledge that our strategy for controlling for the role of financial frictions may be affected by dynamics that are difficult to capture completely in a simple VAR approach. Nonetheless, we present these results as preliminary evidence in favor of an important role for financial frictions in the dynamics of investment and consumption following an uncertainty shock. If availability of credit is indeed a relevant channel that increases the persistence of uncertainty effects in emerging markets, fiscal and monetary

policy actions that soften the credit constraints faced by firms and households may be effective policy tools.

The role played by financial frictions is one candidate – a leading one in our opinion – to explain the differences in the response of emerging markets and developed economies to uncertainty shocks. Nonetheless, other factors may play an important role in explaining these differences, such as labor market rigidities or procyclical macroeconomic policy frameworks. As documented by Kaminsky et al. (2005), the procyclicality of macroeconomic policy in emerging market economies has played a crucial role in explaining their high macroeconomic volatility. Nonetheless, the countercyclical stance of macroeconomic policy in many emerging market economies in response to the global financial crisis of 2008–2009 indicates that things may be changing. All these factors are worth exploring in future research.

4.4. Robustness checks

4.4.1. Uncertainty shock variable

Alternative methodologies are available for the construction of the uncertainty shock variable, even if the data source and shock identification criteria are maintained. In Bloom (2009), the benchmark model includes a binary uncertainty shock variable that takes the value of 1 during the month where the volatility index reaches its maximum during an episode. This differs from our threshold measure in two ways: (i) our variable is weighted by the magnitude of each shock, and (ii) our variable maintains a positive value for each period in which volatility remains above the threshold.

Bloom (2009)'s estimation sample did not include the 2008–09 crisis. We believe that the persistent and very large rise in volatility generated by this event make it substantively different from previous shocks, and that these differences should be captured in the estimation. Since one of the central results we have presented regards the persistence of falls in activity following uncertainty shocks in emerging countries, it is important to verify that this result is not induced by our definition of the uncertainty variable.

Fig. 5 displays the response of investment and consumption in emerging markets estimated using alternative definitions of the uncertainty shock variable. The solid line corresponds to our benchmark results, which were reported in Fig. 3. The others correspond to response functions from shocks to alternative definitions of the uncertainty shock variable: (S1) a binary variable that takes on a value of 1 for each quarter that volatility exceeds the threshold; (S2) a truncated variable where only the first quarter of a volatility shock takes on a positive value, as in Bloom (2009); (S3) a binary variable where only the first quarter of a volatility shock takes on a positive value; and (S4) the baseline variable, but with the 1997 Asian crisis included in an ad-hoc fashion. Neither the increased amplitude nor duration of reaction functions estimated for emerging markets can be attributed to our methodology for constructing the uncertainty variable.

4.4.2. Alternative specifications

One point of departure between our model and Bloom's (2009) is that we have included fewer regressors in the specification. This ensures that we can maintain a broad sample of countries over a long enough sample period to estimate the models. To check the robustness of our results, we estimate the following alternative specifications: (R1) includes the unemployment rate as an additional endogenous variable; (R2) includes the real ex-post overnight interest rate as an additional endogenous variable; (R3) uses an alternative ordering of our benchmark specification, placing quantities before prices as endogenous variables; and (R4) replaces the exogenous block with local stock variables as described below.

Volatility has been shown to be highly synchronized across OECD countries due to spillovers between integrated financial markets. As Gourio et al. (2013) point out, this justifies the use of a common global uncertainty measure. In emerging markets, however, it is possible that

¹³ The Philippines has by far the worst institutions in our sample as reported by the World Bank's Doing Business, and is ranked 141st out of the 181 countries surveyed.

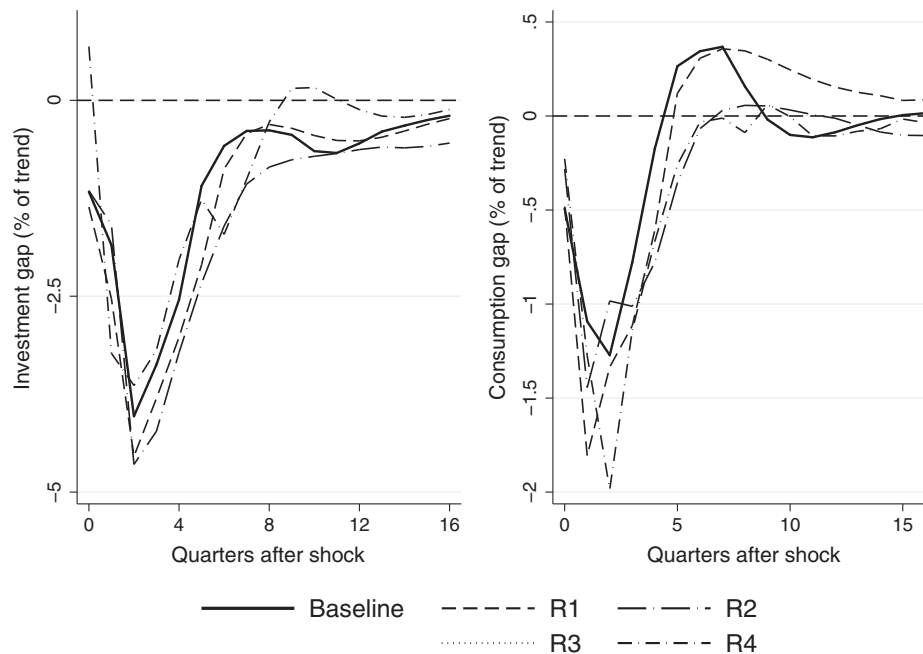


Fig. 7. Impulse-response functions following an uncertainty shock using alternative model specifications.

idiosyncratic or regional events may not perfectly correlate with series generated from developed financial markets. If that is the case, the series we have used so far will be inadequate for capturing local market uncertainty. For instance, the Asian Crisis of 1997 and the Tequila Crisis of 1995 had large impacts on real variables in the Far East and Latin America, respectively, but are not identified as relevant events in the U.S. series. It may also be the case that our orthogonalization to U.S. stock market performance is not effectively separating the effect of uncertainty from local first-moment shocks.

To test the sensitivity of our results to this critique, we use national stock market indices to identify local uncertainty shocks by applying the same identification algorithm as described in Section 2 for all emerging countries in our core sample.¹⁴ While the Russian and 2008–09 financial crises are identified in all the local series, the Enron scandal is not identified as an uncertainty shock in any of them. Fig. 6 shows the local volatility series and identified shocks in Hong Kong's Hang Seng Index and Chile's IPSA index. As expected, the 1997 Asian Crisis figures prominently as a major event in Hong Kong, as does the Tequila Crisis of January 1995 in the Chilean market.

In specification (R4), we relax the small open economy assumption such that the local stock return is treated as an endogenous variable along with local macroeconomic quantities:

$$\begin{aligned}\tilde{s}_t &= \alpha_1 \tilde{s}_{t-1} + \alpha_2 \tilde{s}_{t-2} + \alpha_3 vol_{t-1} + \alpha_4 vol_{t-2} + \gamma_1 (L + L^2) \mathbf{X}_t + \varepsilon_t^s \\ vol_t &= \phi_1 \tilde{s}_{t-1} + \phi_2 \tilde{s}_{t-2} + \phi_3 vol_{t-1} + \phi_4 vol_{t-2} + \varepsilon_t^{vol}\end{aligned}$$

The remaining block of endogenous variables is unchanged from Eqs. (3) through (5).

Fig. 7 shows the benchmark median impulse-response function for the investment and consumption gaps in the core sample, alongside the same functions estimated in the four robustness models discussed in this section. The dynamics are roughly similar on the investment side, and the amplitude of the fall in private consumption increases to 2% of trend. The dynamics of both are roughly similar, however, suggesting that the use of a global series is sufficient for

capturing the response to uncertainty shocks in emerging markets, while recognizing that local uncertainty may further aggravate their effect on demand.¹⁵

5. Conclusion

We have examined to what extent the stylized facts reported in the recent literature on uncertainty shocks are applicable to a heterogeneous group of countries. We identify three findings about the impact of a global uncertainty shock: (i) emerging market countries suffer a median fall in investment approximately four times as large as that found in developed countries; (ii) there is considerable heterogeneity in recovery times between countries and, on average, the recovery time to such a shock is much longer for emerging markets; and (iii) emerging markets, unlike their developed counterparts, experience a strong fall in private consumption following the uncertainty shock. Whereas there is no significant reaction of private consumption to uncertainty shocks in developed economies, emerging economies display a large drop of this variable.

We explore alternative explanations and discuss their policy implications, but our empirical strategy is not equipped to identify the relative importance of each channel in generating the cross-country heterogeneity. Certain structural parameters in a real-option value model can be adjusted to account for the cross-country differences, such as the stochastic discount factor and the degree of irreversibility of investment. However, identifying these deep parameters empirically is difficult and reproducing the degree of heterogeneity we have estimated would thus involve substantial calibration, with questionable implications for policy.

We present preliminary evidence that the amplitude of the drop in investment is correlated with the quality of business institutions, the depth of the local financial sector, and the degree of financial dollarization. These empirical findings suggest that financial frictions may have important interactions with the transmission of global uncertainty shocks, which has implications for the modeling strategy

¹⁴ In the early 1990s, many emerging market stock indices are highly volatile, reflecting very small volumes. We estimate the model with local shocks for the period of 1997:1–2011:1.

¹⁵ We also estimate the model using local shocks identified using a local volatility series that has been orthogonalized to remove movements in global uncertainty, using the residuals from a linear regression of vol_t on vol_t^g . The results are nearly identical to those reported in Fig. 7.

that should be employed to study these events. We then carry out a counterfactual exercise to examine the importance of the credit channel. While displaying a low degree of statistical significance, our estimates show that the contraction in bank loans following uncertainty shocks can account for as much as half of the increased amplitude of the fall in investment in emerging economies with less-developed financial markets. This suggests that the real options approach employed in the literature may be appropriate for modeling the effect of uncertainty in emerging markets, if a credit channel is included.

Disentangling the importance of each of the channels considered in this paper is of primary relevance to policymakers. The real-option literature of investment under irreversibility and uncertainty concludes that monetary and fiscal policy have very limited effectiveness during periods of uncertainty. At most, policy should be implemented decisively in order to avoid aggravating the situation by introducing policy uncertainty. However, if a credit friction amplifies the fall in investment and stalls the recovery, then there may be an important role for fiscal and monetary authorities to increase demand by lowering the cost of financing and providing liquidity to firms that would seek to invest once uncertainty has returned to normal levels.

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